

FARM MANAGEMENT SYSTEM

3. PROJECT DESIGN PHASE

3.1 Problem-Solution Fit

The Farm Management System addresses the difficulties of managing farmer, farm, crop and buyer information manually. The proposed design replaces paper-based records and scattered files with a centralized Salesforce platform. It aims to improve data accuracy, simplify record management, provide faster access to information and support efficient monitoring of farm activities.

3.2 Proposed Solution

Aspect	Design
Problem	Manual farm records can result in data errors, missing information and difficulty in monitoring activities.
Solution	A Salesforce-based Farm Management System with centralized agricultural records and controlled user access.
Innovation	Digital record management, data validation, object relationships, cloud storage and dashboard-based monitoring.
Impact	Improved data accuracy, easier farm monitoring, faster reporting and better information management.
Business Model	Digital platform for managing farm and agricultural information.
Scalability	Can support multiple farmers, farms, crops and buyers as the organization grows.

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3.3 Four-Layer Architecture

Layer	Components	Purpose
1. Presentation	Salesforce Lightning UI	Provides user interfaces to create, view and manage agricultural record
2. Application	Validation Rules, Relationships and automation	Applies business rules, validates information and connects related records.
3. Data	Farm, Crop, Farmer and Buyer custom objects	Stores structured farm management information in Salesforce Cloud.
4. Analytics	Reports and Dashboards	Converts stored agricultural data into useful reports and visual information.

3.4 Object and Relationship Design

- **Farmer** is connected to **Farm** through a Master-Detail relationship.
- **Farm** is connected to **Crop** through a Lookup relationship.
- **Buyer** is maintained as a separate entity for storing buyer information.
- **Sales** information can be associated with crop and buyer details for transaction management.
- The design separates information into different objects to make data management organized and efficient.

3.5 Design Goals

- Centralize farmer, farm, crop and buyer information.
- Reduce duplicate and incorrect data.

- Provide secure and controlled access to records.
- Make farm information easy to monitor through reports and dashboards.
- Provide a scalable structure for future agricultural features and integrations